

Project Status Report

Project Name: Detecting Disease Contacts

Team Name: Circuit Breakers

Project Manager: Robin Pottathuparambil

Team Members: Tarod Anderson, Jared Beyers, Ronil Doshi, Isaiah Maberry, Jennifer Tapia

Report Date: October 9, 2025

Reporting Period: September 12, 2025 to October 9, 2025

Management Summary

Defined milestones completed: 8 of 13 (61.54%)
Defined tasks completed: 0 of 0 (0%) of child tasks in work breakdown structure
Total estimated project hours used: 600 of 1600 hours (37.5%)
Ahead of (or Behind) schedule by: 0 labor hours, 0 days
Known defects: N/A
Staff members on project: N/A
Contingency hours remaining: 100% of 350 hours

Schedule

Initial estimated completion date: *April 9, 2026*
Previous estimated completion date: *N/A*
Current estimated completion date: *April 9, 2026*

Key Milestones Table

ID	Title	Planned Completion Date	Previous Forecast Completion Date	Current Forecast Completion Date	Actual Completion Date
1	<i>Formed Team</i>	8/21/25	8/21/25	8/21/25	8/21/25
2	<i>Selected Project</i>	8/28/25	N/A	8/28/25	8/28/25
3	<i>Drafted Functional Requirements</i>	9/12/2025	N/A	9/12/2025	10/01/2025
4	<i>Drafted Non-Functional Requirements</i>	9/12/2025	N/A	9/12/2025	10/01/2025
5	<i>Status Report 1</i>	9/11/2025	N/A	9/11/2025	9/11/2025
6	<i>Established an initial budget</i>	9/11/2025	N/A	9/11/2025	9/11/2025
7	<i>Drafted Specifications</i>	10/2/2025	N/A	10/2/2025	10/02/2025
8	<i>Finalized Requirements</i>	10/2/2025	N/A	10/2/2025	
9	<i>Preliminary Design</i>	10/3/2025	N/A	10/3/2025	10/03/2025
10	<i>Parts list</i>	10/9/2025	N/A	10/10/2025	

11	Detailed Design	10/16/2025	N/A	10/16/2025	
12	Testing and Verification	4/1/2025	N/A	4/1/2025	
13	Product Complete	04/9/2026	N/A	4/9/2026	

Product Size

Potential Microcontroller suggestion: Raspberry Pi 5 + Raspberry Pi Zero

Subsystem dimensions: 7x15x2 cm

Central System dimensions: 10x20x5 cm

RAM: 8 GB

Effort

Life Cycle Activity	This Reporting Period (labor-hours)		Project to Date (labor-hours)	
	Planned Effort	Actual Effort	Planned Effort	Actual Effort
Requirements	80	100	100	120
Specifications	60	40	80	60
Preliminary Design	140	140	140	140
Parts Order	100	100	100	100

Cost

Life Cycle Activity	This Reporting Period		Project to Date	
	Planned Cost	Actual Cost	Planned Cost	Actual Cost
Materials	900	0	900	0
Cloud computing	100	0	100	0

Requirements Status

I D	Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
F R 1	The system should operate in both day and night conditions.	Jared Beyers	20%	Testing	0 %
F R 2	The system should trigger the camera system when it detects possible human presence.	Jared Beyers	30%	Testing	0 %

F R 3	The system should use computer vision to determine if a possible presence is a human.	Ronil Doshi	25%	Testing	5 %
F R 4	The system should algorithmically determine and record the closest distance for each contact.	Ronil Doshi	30%	Analysis	5 %
F R 5	The system should detect all humans exiting 2 meter radius.	Jared Beyers	30%	Testing	0 %
F R 6	The system should track all human subjects so long as they remain in the 2 meter radius.	Ronil Doshi	35%	Testing	5 %
F R 7	The system should record GPS location for all encounters.	Isaiah Maberry	20%	Testing	0 %
F R 8	The cloud should calculate and store data based on frequency of contacts.	Tarod Anderson	15%	Demonstration	0 %
F R 9	The system should send data to the cloud for storing and access.	Tarod Anderson	25%	Demonstration	0 %
F R 10	The smartphone app should access data from the cloud.	Tarod Anderson	25%	Demonstration	0 %
F R 11	The smartphone app should display all contacts with their respective timestamps and locations on a map, along with the total number of contacts.	Jennifer Tapia	25%	Demonstration	0 %
F R 12	The smartphone app should have a searchable list of contacts. With each contact showing data on duration, distance, timestamp, and location.	Jennifer Tapia	30%	Demonstration	0 %

FR13	The smartphone app should display frequency of contacts, with filters such as per hour, per day, and per week.	Jennifer Tapia	20%	Demonstration	0 %
FR14	The smartphone app shall interact with cloud database to secure user logins and protect access to contact data.	Isaiah Maberry	20%	Testing	0 %
FR15	The system should delete all user video after processing is complete.	Ronil Doshi	10%	Demonstration	0 %
FR16	The smartphone app should display power levels of each subsystem and the central system.	Isaiah Maberry	15%	Inspection	0 %
FR17	The subsystem should send data to the central system via wifi.	Isaiah Maberry	15%	Inspection	0 %

ID	Non-Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
NFR1	The system should be embedded into a wearable vest.	Tarod Anderson	25%	Inspection	0 %
NFR2	The system should be discrete.	Tarod Anderson	10%	Inspection	0 %
NFR3	The system should use a battery for 12-hour operation.	Isaiah Maberry	10%	Analysis	0 %
NFR4	The system should be under 1 KG.	Ronil Doshi	10%	Inspection	0 %

N FR 5	The system shall be implemented on a PCB that ensures compact size and reliable signal routing.	Isaiah Maberry	20%	Inspection	0 %
N FR 6	All components shall be housed in enclosures to ensure protection and user comfort.	Jared Beyers	20%	Inspection	0%
N FR 7	The smartphone app should be professional, elegant, and user friendly.	Jennifer Tapia	25%	Inspection	0%

Top Five Risks*N/A***Open Issues***N/A***Action Items**

- Finalize Specifications
- Submit Parts and Order
- Create Detailed Design Document
- Purchase 4x Pi0 1x Raspberry Pi 5
- Begin training computer vision module

Defects*N/A*